CO2. AT. Analytical problem.

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35/50

1) Grey Level Transformation

a) $S = T(r)$

for Negative Transformation

$\boxed{S = 255 - r}$

b)

Given,

$r = \{50, 100, 50, 200\}$

Calculation

$S = 255 - 50 = 205$

$S = 255 - 100 = 155$

$S = 255 - 150 = 105$

$S = 255 - 200 = 55$

2) Histogram Equalization

Given,

Intensity	frequency
0	4
1	6
2	10
3	6

Total pixels

$N = 4+6+10+6 = 26$

CDF :

$CDF(0) = \frac{4}{26} = 0.154$

$CDF(1) = \frac{4+6}{26} = \frac{10}{26} = 0.385$

$CDF(2) = \frac{4+6+10}{26} = \frac{20}{26} = 0.789$

$CDF(3) = \frac{26}{26} = 1$

Equalation formula

$s = (L=1) \times CDF(r)$

for 4 gray levels,

$L = 4$

$s = 3 \times CDF$

$s(0) = 3(0.154) = 0.462 \approx 0$

$s(1) = 3(0.386) = 1.155 \approx 1$

$s(0) = 3(0.769) = 2.307 \approx 2$

$s(3) = 3(1) = 3$

equalized Intensities

0, 1, 2, 3

... filtering is preferred because it gives more Important to pixels near the center and less Importance to pixels farther from the center

○ Averaging filtering is given equal Importance to all pixels, whereas Gaussian filtering uses weighted values.

b) $G = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Image :-

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Calculation :-

$$= \frac{1}{16}\{(1\times10)+(2\times20)+(1\times10)+(2\times20)+(4\times40) + (2\times20)+(1\times10)+(2\times20)+(1\times10)$$

$$= \frac{360}{16} = 22.5 = \boxed{23}$$ ∴ New center pix

7)

a) Edge detection is a technique used to Identify boundaries between different regions in an Image.

b) $G_y = \begin{bmatrix} \cdot & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ $P = \begin{bmatrix} p_1 & p_2 & \\ p_4 & p_5 & p_6 \\ p_7 & p_8 & p_9 \end{bmatrix}$

$G_y = (-1P_1) + (-2P_2) + (-1P_3) + (0P_4) + (0P_5) + (0P_6) +$
$(1P_7) + (2P_8) + (1P_9)$

Therefore

$$\boxed{G_y = -P_1 - 2P_2 - P_3 + P_7 + 2P_8 + P_9}$$

5) i) Contrast stretching is used to Increase the range of Intensity values in an Image. It Improves the contrast and makes the Image details Clearer.

b) $$s = \left(\frac{r - r_{min}}{r_{max} - r_{min}}\right) \times 255$$

Given,

$r_{min} = 50$, $r_{max} = 150$

for $r = 50$;

$$s = \frac{100-50}{150-50} \times 255$$

$$= \frac{50}{100} \times 255 = 127.5 \approx 128$$

for $r = 150$;

$$s = \frac{150-50}{150-50} \times 255 = \frac{100}{100} \times 255 = 255$$

Transform value

Original	Transformation
50	0
100	128
150	255

The Interpret Improvement in (50-110, 100-128, 150-255).